

Employee Directory Report

Name: ZUO BOYU | Matric Number: A24CS4045

Working UI Screenshot

The application displays the add/edit form, search and sort controls, seeded employee records, active headcount, and active/inactive status badges.

HUMAN RESOURCES

Employee Directory

Manage staff records with Vue, Axios, Express and MySQL.

7
Total

6
Active

1
Inactive

NEW RECORD

Add Employee

Employee ID

EMP001

Full Name

Aina Sofia Rahman

Email

name@company.my

Department

Select department

Position

Systems Analyst

Hire Date

dd - yyyy

Salary (RM)

1500

☒ Active employee

Add Employee

Search	Sort by	Order	
Name, ID, email or department	Name	Ascending	Reset

EMPLOYEE	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
Aina Sofia Rahman EMP001 - aina.rahman@company.my	HR	HR Executive	2021-03-15	RM 4,200.00	Active	Edit Delete
Daniel Lim Wei Jian EMP002 - daniel.lim@company.my	IT	Systems Analyst	2020-08-01	RM 6,800.00	Active	Edit Delete
Farid Iskandar EMP006 - farid.iskandar@company.my	IT	Frontend Developer	2023-06-05	RM 5,100.00	Active	Edit Delete
Mei Ling Tan EMP005 - mei.tan@company.my	Marketing	Digital Marketing Lead	2022-01-17	RM 6,300.00	Active	Edit Delete
Nur Izzati Hassan EMP003 - izzati.hassan@company.my	Finance	Accountant	2019-11-20	RM 5,900.00	Active	Edit Delete

Figure 1: Vue Employee Directory running at <http://127.0.0.1:5174> with MySQL seed data.

Chapter 8 Learning Outcomes

- **LO1: Connect Vue to a backend.** The Vue app calls the Express REST API on port 3001 and keeps the employee list in the parent component state.
- **LO2: Use Axios.** All HTTP requests go through one Axios instance in `src/services/api.js`, including a request logger and a response error mapper.
- **LO3: Build validated Vue forms.** `EmployeeForm.vue` uses `v-model.trim` and `v-model.number`, validates all required fields, and displays inline errors before any request is sent.
- **LO4: Perform CRUD with MySQL.** Express implements create, read, update, delete, server-side search, and whitelisted sorting with prepared statements through the `mysql2` promise pool.

Implementation Notes

The application is separated into form, list, and search/sort components. The SQL script creates the database, table, and seven sample records across HR, IT, Finance, Marketing, and Operations, with one inactive employee.

Challenges and Resolutions

- **Safe sorting:** Directly accepting a column name from the client would risk SQL injection. The API maps requested sort keys to a fixed whitelist before building the `ORDER BY` clause.
- **Consistent errors:** Network and server errors can have different Axios shapes. The response interceptor converts them into one message format used by the UI banner.
- **Responsive layout:** The form and directory use CSS grid and a horizontally safe table layout so the page remains usable on laptop and tablet widths.